

COMP3234 Written Assignment 2

Question 1

(a)

Destination Address Range	Link Interface
11100000 00***** ***** *****	0
11100000 01000000 ***** *****	1
1110000* ***** ***** *****	2
11100001 1***** ***** *****	3
otherwise	3

- (b)
- Prefix match for the first address is the 5th entry : link interface 3
- Prefix match for the second address is the 3rd entry : link interface 2
- Prefix match for the third address is the 4th entry: link interface 3

Question 2

- (a)
- 168.0.0.16
- (b)
- Four equal size subnets:
168.0.0.16/30
168.0.0.20/30
168.0.0.24/30
168.0.0.28/30

Question 3

Step	N'	D(t),p(t)	D(u),p(u)	D(v),p(v)	D(w),p(w)	D(y),p(y)	D(z),p(z)
0	x	$\emptyset$	$\emptyset$	3,x	b,x	b,x	8,x
1	xv	7,v	b,v		b,x	b,x	8,x
2	xvw	7,v	b,v			b,x	8,x
3	xvwy	7,v	b,v				8,x
4	xvwyu	7,v					8,x
5	xvwyut						8,x
6	xvwyutz						

Question 4

The distance table entries at node z:

		cost to				
		u	v	x	y	z
from	v	∞	∞	∞	∞	∞
	x	∞	∞	∞	∞	∞
	z	∞	6	2	∞	0

		cost to				
		u	v	x	y	z
from	v	1	0	3	∞	6
	x	∞	3	0	3	2
	z	7	5	2	5	0

		cost to				
		u	v	x	y	z
from	v	1	0	3	3	5
	x	4	3	0	3	2
	z	6	5	2	5	0

		cost to				
		u	v	x	y	z
from	v	1	0	3	3	5
	x	4	3	0	3	2
	z	6	5	2	5	0

Question 5

(a) To compute the Internet checksum, we add up the values at 16-bit quantities:

$$\begin{array}{r} 00000001 \quad 00000010 \\ 00000011 \quad 00000100 \\ + 00000101 \quad 00000110 \\ \hline 00001001 \quad 00001100 \end{array}$$

the sum = 00001001 00001100;
the internet checksum required is 11110110 11110011

(b) To compute the Internet checksum, we add up the values at 16-bit quantities:

$$\begin{array}{r} 01000011 \quad 01000100 \\ 01000101 \quad 01000110 \\ + 01000111 \quad 01001000 \\ \hline 11001111 \quad 11010010 \end{array}$$

the sum = 11001111 11010010
the internet checksum required is 00110000 00101101

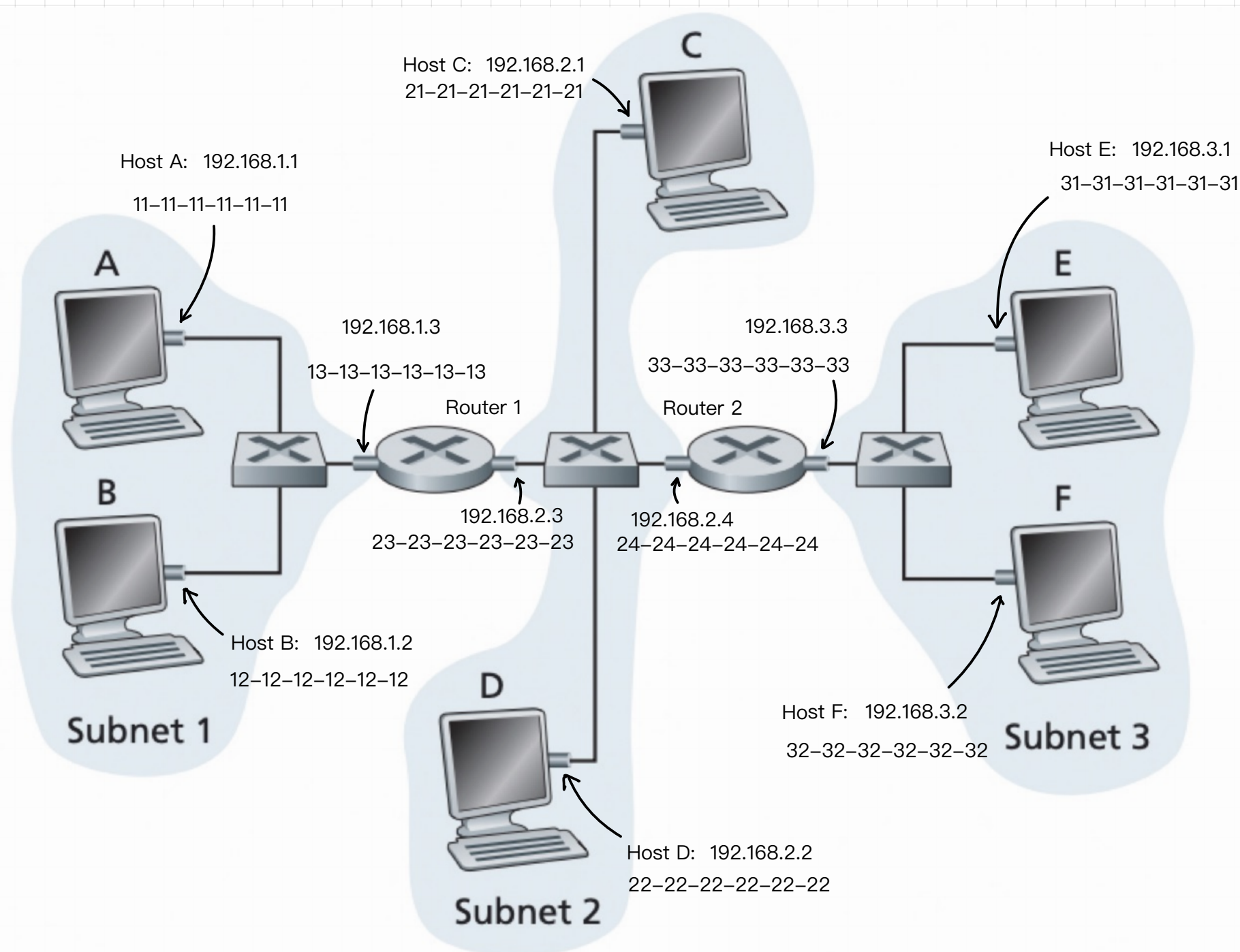
(c) To compute the Internet checksum, we add up the values at 16-bit quantities:

$$\begin{array}{r} 01100011 \quad 01100100 \\ 01100101 \quad 01100110 \\ + 01100111 \quad 01101000 \\ \hline 100110000 \quad 00110010 \\ \\ 00110000 \quad 00110010 \\ + \quad \quad \quad 1 \\ \hline 00110000 \quad 00110011 \end{array}$$

the sum = 00110000 00110011
the internet checksum required is 11001111 11001100

Question 6

For part (a) & (b), please refer to the diagram below:



- (c) 1. In host E, a datagram is created with source IP address 192.168.3.1 and destination address 192.168.1.2
2. The forwarding table in host E shows that the datagram should be sent to interface 192.168.3.3
3. The adapter in host E creates a frame with destination MAC address 33-33-33-33-33-33 and the datagram
4. Router 2 receives the packet and extracts the datagram. The forwarding table in this router shows that the datagram is to be sent to 198.162.2.3
5. Router 2 then creates and sends a packet with the destination MAC address of 23-23-23-23-23-23 and source MAC address of 24-24-24-24-24-24 via its interface with IP address of 198.162.2.4
6. Router 1 receives the packet and extracts the datagram. The forwarding table in this router shows that the datagram is to be sent to 198.162.1.2
7. Router 1 then creates and sends a frame with the destination MAC address of 12-12-12-12-12-12 and source MAC address of 13-13-13-13-13-13 via the interface with IP address of 198.162.1.3
8. Host B receives the packet and extracts the datagram. The process finishes as the packet has reached Host B.